

Model-Reported Uncertainty Finds Hard Items, Not Model Errors: LLM Annotators Under Human Disagreement and the Economics of Routing

Yogya Mehrotra

yogyamehrotra@gmail.com

github.com/yogyam/DisagreeBench

Abstract

Claims that LLMs can replace human annotators rest on aggregate agreement numbers that average over a bimodal reality: most items are easy, and a contested minority is not. On ChaosNLI (3,113 NLI items with 100 annotations each), claude-sonnet-5’s divergence from the human label distribution quadruples across human-entropy quintiles (JSD $0.07 \rightarrow 0.31$) while accuracy against the majority label declines only gently ($0.94 \rightarrow 0.56$) — the standard metric flatters the model precisely on contested items. The obvious repair — route items the model is unsure about to humans — fails at the first step: the model answers identically across 10 samples on 85% of items (Krippendorff’s $\alpha = 0.90$ vs. 0.37 for the human pool), and its sampling entropy barely correlates with human disagreement ($\rho = 0.164$); enabling reasoning does not change this. Asked instead to *estimate* the annotator distribution, the same model produces distributions $3\times$ closer (JSD 0.07) whose entropy does clear the routing threshold ($\rho = 0.393$). A budget-sweep router simulation with regret oracles quantifies the stakes: with sampled-distribution fallbacks, verbalized-entropy routing captures 37% of the achievable oracle gain and random allocation needs $\sim 1.5\times$ the budget to match it; against the model’s own verbalized fallback, the same signal is *worse than random* — residual errors hide in confidently-wrong low-entropy items. These findings replicate on conversational safety (DICES-350) and on LeWiDi-2025 labels that post-date the model’s training, and a split-half noise-ceiling analysis shows per-item routing gates are only measurable against deep annotator panels. Model-reported uncertainty finds hard items, not model errors.

1 Motivation

Data-labeling providers are being asked to justify themselves against claims of the form “the model agrees with human labels 90% of the time

at 1% of the cost.” That figure is an average over a distribution of examples that is sharply bimodal in difficulty: most items in a real annotation stream are unambiguous, while a minority are genuinely contested — and the contested minority consumes most of a real annotation budget and drives most disagreement in downstream training. If LLM agreement concentrates on the easy majority, a headline agreement figure can be technically true and operationally misleading, and the correct architecture is neither “replace humans” nor “keep humans” but *triage*. This study measures the premise of that argument directly, and then tests whether the triage router is buildable from the model’s own uncertainty.

2 Data

ChaosNLI (Nie et al., 2020) provides ~ 100 annotations per item for 4,645 development-set items from SNLI, MNLI, and Abductive NLI. We use the SNLI (1,514) and MNLI (1,599) portions — 3,113 items sharing the 3-way entailment/neutral/contradiction label space — and exclude Abductive NLI (a 2-way task). The 100-annotator sample gives a genuine label distribution per item rather than a noisy 3–5-rater estimate, which is what makes entropy stratification statistically meaningful. Items are bucketed into quintiles of human label entropy computed from the full distribution (quintile edges: 0.66, 0.91, 1.05, 1.21 bits).

The pool is genuinely contested: nominal Krippendorff’s α is 0.447 (SNLI), 0.283 (MNLI), 0.367 combined. This is by construction — ChaosNLI drew items whose original 5-rater annotations disagreed — and it is exactly the regime the substitution argument is quietly averaged over.

3 Method

Elicitation. claude-sonnet-5 (Anthropic), accessed via the Message Batches API. Each item is sampled 10 times as an independent request. The label options are presented in randomized order per sample (mitigating option-ordering bias); output is constrained to a JSON schema over the three labels, so every sample parses; extended thinking is disabled. The empirical distribution over the 10 sampled labels is the model’s label distribution. Of 31,130 requests, 31,127 returned usable labels (3 errors).

Verbalized arm. As a comparison elicitation (one request per item, full 3,113-item set), the model is asked to estimate how 100 careful annotators would distribute over the three labels, output as integer percentages under the same schema constraint.

Router simulation. Each item’s 100 annotations are split 50/50: one half defines the ground-truth distribution, the other is the finite pool routed items draw from (so the router can never “buy” the evaluation target, and even perfect routing pays annotator-sampling noise). A router scores all items, sends the top fraction f to humans — receiving k annotations drawn without replacement from the pool — and keeps the model’s distribution for the rest. We sweep f from 0 to 1 and $k \in \{1, 3, 5, 10, 20, 50\}$, with 20 replicate splits. Strategies: random; self-consistency entropy; verbalized entropy; channel disagreement (JSD between the sampled and verbalized distributions); an *entropy oracle* scoring by true human entropy; and two regret oracles scoring by each item’s actual improvement from routing — *expected gain* (Monte-Carlo mean over 200 re-draws; the fair upper bound, since it knows which items benefit but not the luck of the draw) and *realized gain* (a skyline that peeks at the draw). Headline numbers: AUC of the budget–JSD curve, normalized as the share of the expected-gain oracle’s improvement over random (“capture”); and budget-to-match, the multiple of annotations random allocation needs to reach a strategy’s JSD at a given budget. Item-level bootstrap (1,000 resamples) gives CIs on strategy–random gaps. Everything is repeated with total variation distance in place of JSD and with a bootstrap-resampling split protocol in place of the 50/50 split.

Quintile	n	Mean JSD [CI]	Acc. vs. maj. [CI]
Q1 (low)	631	0.074 [.064, .085]	0.945 [.926, .962]
Q2	615	0.178 [.165, .192]	0.805 [.772, .836]
Q3	623	0.244 [.232, .255]	0.674 [.637, .711]
Q4	622	0.263 [.254, .273]	0.614 [.576, .653]
Q5 (high)	622	0.311 [.302, .321]	0.558 [.519, .596]

Table 1: Model–human divergence and accuracy against the majority label, by quintile of human label entropy (95% bootstrap CIs).

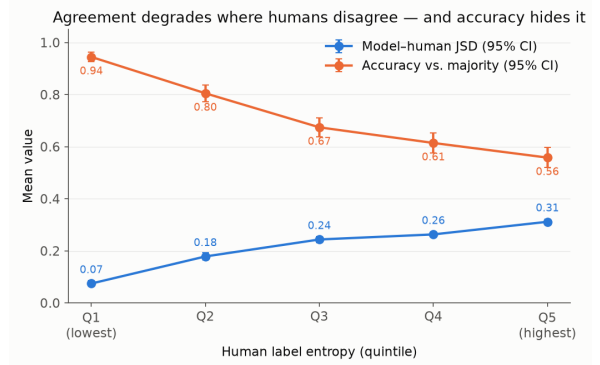


Figure 1: Mean JSD between model and human label distributions (blue) and accuracy against the majority label (orange) across human-entropy quintiles. The divergence quadruples while accuracy declines gently: the standard metric is most flattering exactly where the model is furthest from the human distribution.

Metrics. Primary: per-item Jensen–Shannon divergence (base 2) between the model and human label distributions, averaged per entropy quintile. Reported alongside, explicitly as the metric under critique: accuracy of the model’s modal label against the human majority label. The routing gate: Spearman rank correlation between model self-consistency entropy and human entropy. All means and the correlation carry 95% percentile bootstrap CIs (10,000 resamples). Contamination diagnostic: agreement of the modal label with the *original* single gold label vs. with the ChaosNLI majority, per quintile.

4 Results

4.1 Agreement degrades with human disagreement — and accuracy hides it

JSD rises monotonically by a factor of ~ 4.2 from Q1 to Q5 (Table 1, Figure 1). This deliberately replicates the stratified diagnostic Níe et al. (2020) ran on fine-tuned encoders, and the shape has changed in a way that matters: where their models’ accuracy fell to chance on contested items, the frontier LLM stays well above it, and its Q1

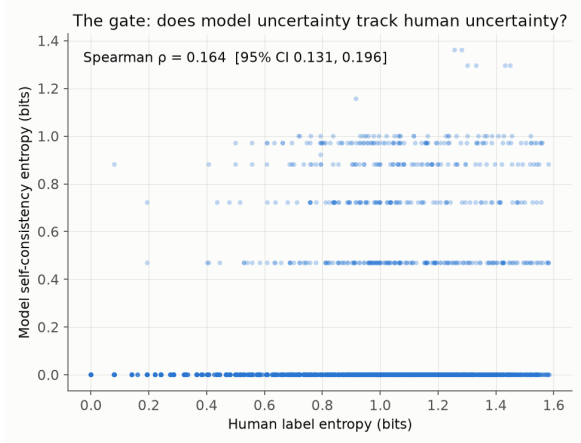


Figure 2: Model self-consistency entropy vs. human label entropy. The model sits at zero entropy on 85% of items regardless of how contested they are; $\rho = 0.164$.

JSD now reaches their estimated human bound (~ 0.06). Accuracy against the majority declines much more slowly, and its floor is deceptive: on Q5 items the human majority is itself close to a coin flip, so “56% accuracy” largely reflects picking the more common side of a split — not agreement with what humans collectively believe. The gap between the two curves is the headline finding: **the standard metric is most flattering exactly where the model is furthest from the human distribution.**

4.2 The gate fails: model uncertainty is not human uncertainty

If a model knew which items humans find hard, its uncertainty could route those items to people. It does not. Across 10 samples the model returns the identical label on 2,653 / 3,113 items (85%) — including 84% of Q5, where human annotators split at more than 1.2 bits of entropy. Where the human pool agrees at $\alpha = 0.367$, the model agrees with itself at $\alpha = 0.904$.

The rank correlation between model self-consistency entropy and human entropy is $\rho = 0.164$ [95% CI 0.131, 0.196] ($p \approx 4 \times 10^{-20}$; Figure 2). The correlation is reliably nonzero — the model is not blind to difficulty — but the CI excludes by a wide margin any threshold ($\geq \sim 0.3$) at which uncertainty-based routing would meaningfully beat random allocation. An ablation with adaptive thinking enabled (§6) leaves the picture unchanged. By the pre-registered logic of the study design, self-consistency entropy is disqualified as a router input: **this routing signal does not**

Signal	JSD	Entropy (bits)	Gate ρ
Sampled ($10\times$)	0.224	0.089	0.164
Verbalized	0.072	0.863	0.393
Human pool	—	0.939	—

Table 2: Mean JSD vs. the human distribution, mean entropy, and Spearman correlation with human entropy, for the two elicitation channels (full set, $n = 3,113$; verbalized gate $p \approx 10^{-113}$).

exist (§4.5 confirms end-to-end that routing on it is no better than random). The signal that does exist comes from a different channel entirely (§4.3).

4.3 Verbalized confidence: the model can report disagreement it does not enact

Prompted instead to *estimate* the human distribution directly — “imagine 100 careful annotators; how many choose each label?” (one request per item) — the model behaves entirely differently (Table 2).

Verbalized distributions are $\sim 3\times$ closer to the human distribution than sampled ones, their average entropy nearly matches the human pool’s, and verbalized entropy clears the pre-registered routing threshold ($\rho = 0.393 \geq 0.3$; pilot estimate was 0.354). Where sampled JSD degrades $4.2\times$ from the lowest- to highest-entropy quintile (§4.1), verbalized JSD is nearly flat: $0.044 \rightarrow 0.103$. The declarative estimate stays accurate on precisely the contested items where the behavioral distribution collapses. Strikingly, on ChaosNLI the two uncertainty signals are essentially uncorrelated with each other ($\rho = 0.03\text{--}0.08$, depending on which sampled run is paired): they measure different quantities (§4.6 shows this independence is dataset-dependent, though the quality gap is not). Together with §4.2, this yields the paper’s sharpest formulation: **the model’s behavioral uncertainty carries no information about human disagreement, but its declarative estimate of human disagreement does.** This inverts a common assumption (including in our own study design) that verbalized probabilities are poorly calibrated and self-consistency is the trustworthy signal. It also conditionally revives the triage router: the routing signal exists — it just has to be asked for, not observed.

The direction of this contrast — verbalized beats sampled, and reasoning does not close the gap — independently replicates what Ni et al. (2026) report on hate-speech, emotion, and preference tasks and Meister et al. (2025) on opinion

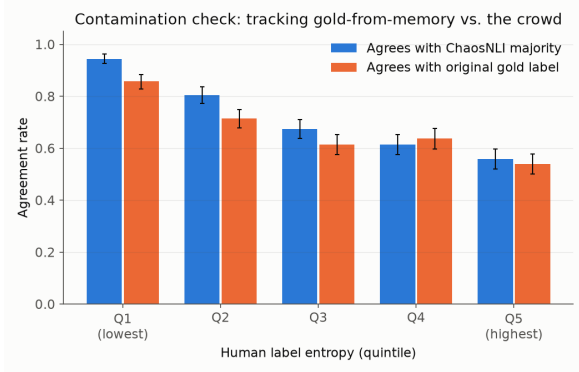


Figure 3: Modal-label agreement with the ChaosNLI 100-annotator majority vs. the original single gold label, by quintile. The model tracks the crowd, not the memorized gold.

surveys; §5.3 details what is new here (the entropy-flatness, the channel independence, and the routing consequences in §4.5).

One caveat attaches specifically to this result: ChaosNLI’s label *distributions* have been public since 2020 — and are by now an explicit optimization target in the literature (e.g., Wu et al., 2026, fine-tunes directly on them) — so the model may have absorbed some of them in pretraining. A memorized distribution would inflate verbalized performance in a way the gold-label diagnostic in §4.4 does not cover. §4.6 supplies the needed check: the verbalized channel’s distributional advantage holds on LeWiDi-2025 paraphrase labels released after the model’s training cutoff.

4.4 The result is not memorization

ChaosNLI is built on SNLI/MNLI, which are almost certainly in the model’s pretraining data; a model could look good by reciting original gold labels. The diagnostic says otherwise: the modal label agrees with the ChaosNLI majority *more* than with the original gold label in Q1–Q3 (e.g. 0.945 vs. 0.857 in Q1), and the two converge in Q4–Q5 (0.614 vs. 0.637; 0.558 vs. 0.539). If the model were reproducing memorized gold labels, the pattern would invert on high-entropy items, where the original gold is essentially arbitrary. It does not (Figure 3).

4.5 The router: uncertainty finds hard items, not model errors

The simulation (method in §3) asks the operational question directly: given a fixed annotation budget, does routing on model-reported uncertainty beat spending the same budget at random? The answer

splits cleanly by what the unrouted items fall back to.

Against the sampled base — the realistic setting where un-routed items keep the model’s self-consistency distribution — verbalized-entropy routing works (Figure 4, left). At $k = 10$ annotations per routed item it captures **37.4%** of the expected-gain oracle’s improvement over random; channel disagreement (JSD between the model’s two distributions, using no human information) captures **45.5%**, statistically indistinguishable from the *true-human-entropy* oracle (47.7%). In budget terms, random allocation needs **1.44–1.57×** as many annotations to match verbalized-entropy routing at 10–30% coverage (strategy–random gap CIs exclude zero at every budget above 5%). Self-consistency entropy captures nothing (−8.3%), closing the loop on §4.2 with an end-to-end null.

Against the verbalized base — where unrouted items keep the model’s verbalized distribution, the stronger baseline §4.3 establishes — the same signal *inverts* (Figure 4, right): verbalized entropy is reliably worse than random (capture −18.7%; random matches it with 0.5–0.6× the budget). The mechanism is anti-selection. High verbalized entropy flags items where humans genuinely disagree — but §4.3 showed the verbalized distribution is already accurate exactly there. The residual errors sit in low-entropy items the model is confidently wrong about, which entropy routing systematically deprioritizes (per-item verbalized entropy vs. verbalized error: $\rho \approx -0.07$). Even the true-human-entropy oracle barely beats random here (11.9%), while the expected-gain oracle still improves substantially — the errors are findable in principle, just not by any entropy-shaped signal. **Model-reported uncertainty tells you where the task is hard, not where the model is wrong**; it routes usefully only when the fallback annotator is weak.

Breadth vs. depth. Sweeping k (Figure 5) shows that at a fixed total budget, shallow-and-broad beats deep-and-narrow on the sampled base ($k = 3$ –5 dominates $k = 20$ –50), and yields the study’s most quotable number on the verbalized base: replacing the model’s verbalized distribution with the empirical distribution of k human labels only breaks even at $k \approx 5$ ($k = 1$: JSD 0.270; $k = 3$: 0.113; $k = 5$: 0.072; model verbalized:

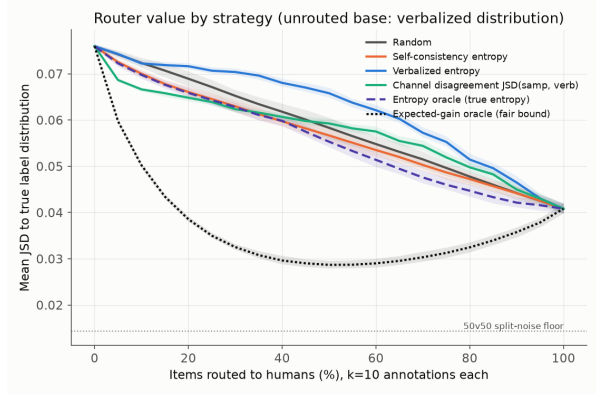
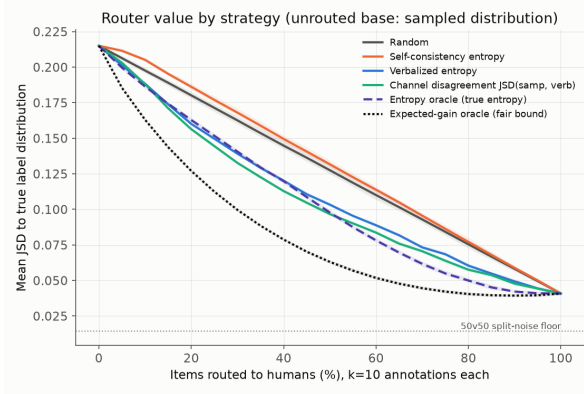


Figure 4: Router value curves at $k = 10$ annotations per routed item. **Left:** unrouted items keep the model’s sampled distribution — verbalized-entropy and channel-disagreement routing approach the true-human-entropy oracle. **Right:** unrouted items keep the model’s verbalized distribution — the same verbalized-entropy signal is now *worse than random* (anti-selection), and even the true-entropy oracle barely beats random; the expected-gain oracle shows the errors remain findable in principle.

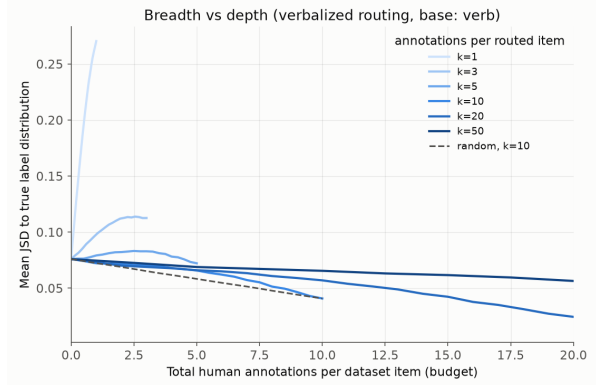
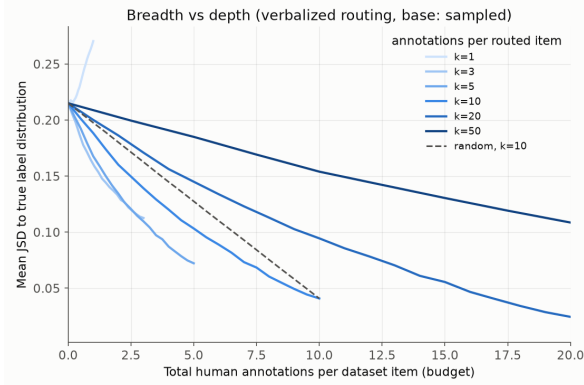


Figure 5: Breadth vs. depth: value of verbalized-entropy routing as a function of total annotation budget, for different annotation depths k per routed item. **Left:** sampled base — shallow-and-broad ($k = 3\text{--}5$) dominates deep-and-narrow. **Right:** verbalized base — replacing the model’s verbalized distribution with the empirical distribution of k human labels only breaks even at $k \approx 5$; at $k = 1$ routing makes the corpus estimate strictly worse.

0.076). In distributional terms, one model call is worth roughly five human annotations on this dataset — with the memorization caveat of §4.3 attached.

All orderings and magnitudes are stable under total variation distance in place of JSD and under the bootstrap split protocol (capture 37.7–38.9% verbalized, 44.0–46.8% channel disagreement on the sampled base; anti-selection persists on the verbalized base in every variant).

4.6 Does it generalize beyond NLI?

We pre-registered the expectation that the channel dissociation is a property of aligned models, not of NLI, and replicated the pipeline on two further datasets: **DICES-350** (Aroyo et al., 2023), 350 conversational-safety items rated Yes/No/Unsure

by the same 104 raters (richer disagreement than ChaosNLI: no item is unanimous), and the **LeWiDi-2025 Paraphrase** set (Leonardelli et al., 2025), 500 question pairs whose labels were first released in May 2025 — after the model’s training — making it the one dataset where memorization is excluded by construction (4-annotator panel, so gate-scope only).

Table 3 summarizes. Every structural finding replicates on safety data: determinism is even more extreme (93.7%), sampled JSD is numerically almost identical (0.225), the verbalized channel is $2.5\times$ closer with near-human mean entropy (0.91 vs. 1.04 bits) and a much flatter difficulty profile ($0.062 \rightarrow 0.107$ across quintiles vs. $0.133 \rightarrow 0.326$ sampled). The router replicates directionally with softer magnitudes: verbalized-entropy

	ChaosNLI	DICES	Par
Determinism (sampled)	0.85	0.94	0.90
JSD sampled	0.224	0.225	0.224
JSD verbalized	0.072	0.090	0.154
Gate ρ (sampled)	0.164	0.129	0.128
Gate ρ (verbalized)	0.393	0.289	0.039 [†]
Split-half ceiling	0.843	0.750	-0.002
Gate / ceiling	0.47	0.39	n/a

Table 3: Cross-dataset replication. [†]The Par gate is *unmeasurable*, not failed: a 4-annotator panel’s split-half entropy correlation is zero, so no signal could register. Gates should be reported relative to the panel’s noise ceiling.

routing captures 28.7% of the expected-gain oracle against the sampled fallback (random needs 1.25–1.30 $\times$ the budget), and the **anti-selection reversal reappears** against the verbalized fallback (capture -11.0%; strategy-random gap CIs exclude zero at 10–30% budgets). On Par, the verbalized channel’s distributional advantage holds (JSD 0.154 vs. 0.224; mean entropy 0.568 vs. human 0.555) on labels the model cannot have seen, which weakens a memorization account of the verbalized results — though the per-item gate cannot be evaluated there at all: our split-half analysis shows a 4-annotator panel’s entropy carries no measurable per-item signal (ceiling -0.002 vs. 0.843 for ChaosNLI’s 100 raters and 0.750 for DICES’s 104). Two honest amendments emerge. First, the verbalized gate is not universal: normalized by the ceiling it declines from 47% of achievable correlation (NLI) to 39% (safety), and its DICES CI [0.19, 0.38] straddles the 0.3 threshold. Second, the channel-*independence* finding is dataset-dependent: sampled and verbalized entropies correlate at $\rho \leq 0.08$ on ChaosNLI but $\rho \approx 0.35$ –0.42 on DICES and Par — the channels are always unequal in quality, but not always uncorrelated.

5 Related Work

5.1 Human label variation and distributional evaluation

That disagreement in NLI judgments is reproducible signal rather than annotation noise was established by Pavlick and Kwiatkowski (2019), and the position that label variation should be modeled rather than adjudicated away is now a program (Plank, 2022; Uma et al., 2021; Basile et al., 2021; Cabitza et al., 2023). Our evaluation toolkit is theirs: JSD against the human label

distribution and the correlation between per-item model and crowd entropy are both standardized in Uma et al. (2021). Our dataset paper, ChaosNLI (Nie et al., 2020), already performed the stratified analysis for fine-tuned encoders: accuracy degrades toward chance and JSD stays above 0.2 as human agreement falls. §4.1 is a deliberate replication of that diagnostic on a frontier LLM, and what changed is the shape: the model now reaches the estimated human JSD bound (~ 0.06) on high-agreement items, and accuracy on contested items no longer collapses to chance (0.56 vs. 0.33) — which makes the flattery problem *worse*, because the standard metric now looks respectable everywhere while distributional divergence quadruples. Distributed NLI (Zhou et al., 2022) formalized the task of predicting human opinion distributions on ChaosNLI: their supervised estimators (MC Dropout, recalibration, distillation) reach JSD ≈ 0.18 –0.19, which our zero-supervision verbalized elicitation beats by $\sim 2.5\times$; their appendix also documents that encoder uncertainty is miscalibrated against human disagreement, a precursor of our gate analysis. Baan et al. (2022) showed on ChaosNLI that calibration to a majority label is incoherent under disagreement (our 50/50 truth/pool split in §3 adapts their device of splitting the 100 annotations into subpopulations); Baan et al. (2024) argue conceptually that a predictive distribution conflates model confidence with human variation — our channel-decorrelation finding (§4.3) turns that distinction into a measured property. VariErr NLI (Weber-Genzel et al., 2024) implies some of ChaosNLI’s variation is annotation error, which makes our router’s measured gains conservative with respect to error-cleaned targets.

5.2 LLMs as annotators

The replacement wave (Gilardi et al., 2023; Törnberg, 2023) reads the model’s high self-consistency as annotation quality; on contested items we show the same property is a liability — the model agrees with itself ($\alpha = 0.90$) about questions on which 100 humans split nearly evenly ($\alpha = 0.37$). Reiss (2023) complained that 2023-era ChatGPT was too *unstable* to annotate reliably; 2026 frontier models exhibit the opposite pathology, and neither regime tracks human disagreement. Baumann et al. (2025) quantify the downstream stakes — LLM annotation choices flip roughly a third of tested conclusions

— and already propose verbalized-confidence-targeted human annotation as a mitigation, evaluated against disagreement-filtered gold labels; our anti-selection result (§4.5) exposes a failure mode of exactly that strategy once the target is the label distribution and the fallback is the model’s own best output.

5.3 LLM uncertainty channels and human disagreement

Lee et al. (2023) first compared sampled and logprob-derived LLM distributions to ChaosNLI’s annotator distributions, observing that GPT-3’s output entropy is near zero and visibly uncorrelated with human entropy. Our §4.2 quantifies that observation at frontier scale (85% exact determinism; $\alpha = 0.90$ vs. 0.37; gate $\rho = 0.164$ with CIs) and stress-tests it against reasoning and paraphrase confounds. Madaan et al. (2025) stratify accuracy by human entropy for open-weight models on ChaosNLI; the accuracy/JSD *dissociation* across quintiles is ours. The channel contrast we find was anticipated on other tasks: Ni et al. (2026) show verbalized distributions beat sampled ones for disagreement prediction on hate-speech, emotion, and preference tasks (and that RLVR-style reasoning hurts); Meister et al. (2025) found LLMs describe opinion distributions better than they simulate them; Jang et al. (2026) attribute the simulation failure to alignment-induced mode collapse (cf. Kirk et al., 2024), consistent with the calibration literature where asking beats token probabilities (Lin et al., 2022; Tian et al., 2023; Kadavath et al., 2022) — though against *correctness*, consistency-based signals often win (Xiong et al., 2024), underlining that our target is different. Wang et al. (2024) show first-token probabilities diverge from what instruction-tuned models actually answer, which is why we omit a logprob arm. On NLI specifically, Chen et al. (2024, 2025) approximate human judgment distributions with explanation-conditioned LLM prompting, and Chen et al. (2026) show CoT does not calibrate distributions on ambiguous items. Relative to all of these, our §4.3–4.4 contribute: replication of the verbalized–sampled gap on 100-way NLI distributions with a frontier model; the near-flatness of verbalized JSD across disagreement levels; the statistical *independence* of the two channels’ entropies on ChaosNLI ($\rho \leq 0.08$; dataset-dependent per §4.6), which no prior work reports; and the gate framing — with its split-half

noise ceiling — that connects channel quality to routability. Zhang et al. (2026) propose verbalized sampling to recover output diversity; our results externally validate the verbal channel against ground-truth human distributions rather than diversity alone.

5.4 Hybrid annotation and budget allocation

Routing between model and human annotators instantiates learning-to-defer (Madras et al., 2018; Mozannar and Sontag, 2020; DeSalvo et al., 2025), but that literature assumes a trained rejector and single-label accuracy; we ask which zero-shot LLM signal could support deferral at all, with distributional fidelity as the objective. CoAnnotating (Li et al., 2023) routes on prompt-paraphrase entropy against single gold labels and reports scalar verbalized confidence unreliable — the opposite ordering from ours, which dissolves once the elicitation target is distinguished: they elicit scalar self-confidence scored against gold labels; we elicit a full annotator distribution scored against real human distributions. HyPAC (Zeng et al., 2026) gives PAC-guaranteed routing for 0-1 error against objective answers; production systems (Kim et al., 2024; Bachar et al., 2026) route escalation on uncertainty and find raw signals wanting. Gligorić et al. (2025) allocate human budget by verbalized confidence for population-level estimates; Mehrotra et al. (2026) do so at demographic-group level; Hakimi et al. (2026) find uncertainty-driven selection fails to beat random in active learning; Schroeder et al. (2025) show humans anchor when *verifying* LLM labels, which motivates our design choice of independent annotation on routed items. Closest to our simulation: Klugmann et al. (2024) route items between crowd-trained soft-label predictors and humans in vision, and Peale et al. (2026) run budget-swept uncertainty-decomposition routing on ChaosNLI with trained classifiers. Kohli (2026) shows on ChaosNLI that annotations needed per item is metric-dependent (distributional metrics saturate near $N = 10$), consistent with our k -sweep; Gruber et al. (2025) pose the annotator-selection question — whether a human or a model provides each label — as an open empirical problem. Recent budget-aware routing systems allocate items to humans and LLMs by feature-based difficulty (Purificato et al., 2026) or learned annotator models (Moriyama et al., 2026), and Kulmizev et al. (2026) document LLM label variation on NLI — none compare elicitation chan-

nels as routing signals, sweep annotation depth, or evaluate against 100-way label distributions.

Concurrent work. While this paper was being finalized, Lail (2026) (posted September 6, 2026; our results and code were public from August 19, 2026) independently ran an expert-escalation budget simulation on the same ChaosNLI items with a per-item annotation half-split and regret-vs-oracle accounting, and found that neither verbalized confidence nor the entropy of the judge’s verbalized distribution beats random escalation — converging, in null form, with our anti-selection result against the verbalized fallback. The two studies are complementary: Lail has no sampling arm (so no channel comparison and no positive routing result), a single fallback, and no annotation-depth sweep; we do not attempt his epistemic/aleatoric decomposition. That two independent designs both find model-reported uncertainty failing to locate the model’s own errors strengthens the warning label in §7. What remains unique to the present study is the fallback-channel reversal — the same uncertainty signal captures 37% of oracle gain against a sampled fallback yet anti-selects against a verbalized one — along with the two-channel dissociation that explains it and the breadth-vs-depth sweep.

6 Limitations

- **Elicitation is not the explanation (ablated).** Because thinking was disabled and output schema-constrained in the main run, we replicated the 200-item pilot with adaptive thinking enabled and a 2,048-token budget. Determinism was unchanged (84% vs. 87%), model self-agreement was unchanged ($\alpha = 0.896$ vs. 0.904), and the gate correlation did not improve ($\rho = 0.066$ $[-0.06, 0.19]$, n.s. at $n = 200$). Notably, the model — free to reason at its own discretion — mostly declined to (mean 21 output tokens/sample vs. 11 without thinking): it does not even *perceive* these items as hard. The determinism result therefore survives its most obvious confound, though we cannot rule out that stronger forced-reasoning prompts would behave differently.
- **One model.** All results use claude-sonnet-5; §4.6 spans three task types (NLI ambiguity, conversational safety, paraphrase subjectivity)

but no second model family. DICES labels (collected 2023, public since then) are not contamination-proof in the way Par’s are, though per-rater CSVs are a less plausible memorization target than ChaosNLI’s widely-replicated distributions.

- **Sampling temperature is not tunable** on this model (the API rejects non-default sampling parameters), so the temperature-sensitivity analysis in the original design is moot; sampling ran at the provider default.
- **10 samples bound measurable entropy.** With 10 draws the minimum nonzero entropy is ~ 0.47 bits, coarsening the model-entropy scale. This attenuates the gate correlation somewhat; it cannot explain 85% exact determinism.
- **The router is a simulation, not a deployment.** Routed items draw annotations from ChaosNLI’s own pool, so annotators are exchangeable with the evaluation target by construction; a real deployment adds annotator drift, and its ground truth would not be a held-out half of 100 labels. The memorization caveat (§4.3) also propagates: if verbalized distributions are partly recalled, both the verbalized base and the routing signal are optimistic. The sampled-base conclusions are less exposed, since the sampled distributions show no memorization signature (§4.4).
- **Prompt sensitivity: conclusions stable, but the sampled signal itself is noisy.** Re-running the pilot under two prompt paraphrases leaves every headline quantity essentially unchanged (determinism 80–87%; mean JSD 0.21–0.23; gate $\rho = 0.09$ –0.12, never approaching 0.3; modal labels agree across prompts on 94% of items). However, *per-item* self-consistency entropy correlates only moderately across paraphrases (Spearman 0.38–0.46) — the little sampling variance that exists is substantially prompt-specific noise rather than a stable property of the item, a further reason it cannot serve as a routing signal.

7 Implications

For the substitution debate, the results assemble into one architecture with a warning label. The

stratified curve (§4.1) says the cheap examples are the ones the model already labels well — consistent with aggressive automation of the easy majority. The failed gate (§4.2) says the obvious triage signal — “let the model flag what it’s unsure about,” read off its sampling behavior — does not exist. The verbalized result (§4.3) supplies the repair: the signal is a *declarative* capability that must be asked for, and the router built on it (§4.5) delivers real budget savings ($\sim 1.5\times$ annotation efficiency) over random allocation in the realistic setting. The warning label is the anti-selection result: the same uncertainty signal is worse than useless for auditing the model’s own best output, because model-reported uncertainty locates task difficulty, not model error. Practically: use verbalized distributions as the machine annotation, route by uncertainty only to decide *which items get humans at all*, and do not expect the model to tell you where it is wrong — error-finding needs an independent signal. Regardless, headline agreement numbers should be reported stratified by inter-annotator agreement as a matter of course.

Reproducibility

All code, elicitation scripts, and figures are available at github.com/yogyam/DisagreeBench. Pipeline: `prepare.py` $\rightarrow$ `elicit.py` $\rightarrow$ `analyze.py`; verbalized arm: `verbalize.py`; router simulation: `router.py` (pure simulation on saved elicitation outputs; no further API calls). Elicitation: claude-sonnet-5, Message Batches API (verbalized full set ran synchronously), 10 samples/item, randomized option order, JSON-schema output, thinking disabled, seed 20260805 for v0–v1 randomization and bootstraps, seed 20260819 for the router simulation, seed 20260912 for the replication elicitation (`prepare_v3.py`, `elicit_v3.py`). Total API cost: $\approx \$30\text{--}36$ ($\sim 44\text{k}$ requests; $\sim 16.5\text{M}$ input / $\sim 0.5\text{M}$ output tokens). DICES-350 is CC BY 4.0; LeWiDi-2025 data is publicly released by the shared task. ChaosNLI data is not redistributed; `prepare.py` regenerates the processed splits from the original release.

References

Lora Aroyo, Alex S. Taylor, Mark Díaz, Christopher M. Homan, Alicia Parrish, Gregory Serapio-

García, Vinodkumar Prabhakaran, and Ding Wang. 2023. [DICES dataset: Diversity in conversational AI evaluation for safety](#). In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*.

Joris Baan, Wilker Aziz, Barbara Plank, and Raquel Fernández. 2022. [Stop measuring calibration when humans disagree](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Joris Baan, Raquel Fernández, Barbara Plank, and Wilker Aziz. 2024. [Interpreting predictive probabilities: Model confidence or human label variation?](#) In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.

Or Bachar, Or Levi, Sardhendu Mishra, Adi Levi, Manpreet Singh Minhas, Justin Miller, Omer Ben-Porat, Eilon Sheerit, and Jonathan Morra. 2026. [LLM performance predictors: Learning when to escalate in hybrid human-AI moderation systems](#). In *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*.

Valerio Basile, Michael Fell, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, Massimo Poesio, and Alexandra Uma. 2021. [We need to consider disagreement in evaluation](#). In *Proceedings of the 1st Workshop on Benchmarking: Past, Present and Future*.

Joachim Baumann, Paul Röttger, Aleksandra Urman, Albert Wendsjö, Flor Miriam Plaza-del Arco, Johannes B. Gruber, and Dirk Hovy. 2025. [Large language model hacking: Quantifying the hidden risks of using LLMs for text annotation](#). *arXiv preprint arXiv:2509.08825*.

Federico Cabitza, Andrea Campagner, and Valerio Basile. 2023. [Toward a perspectivist turn in ground truthing for predictive computing](#). *Proceedings of the AAAI Conference on Artificial Intelligence*, 37.

Beiduo Chen, Tianyi Hu, Chenyang Zhang, Robert Litschko, Anna Korhonen, and Barbara Plank. 2026. [Decoupling the effect of chain-of-thought reasoning: A human label variation perspective](#). In *Findings of the Association for Computational Linguistics: 2026*.

Beiduo Chen, Siyao Peng, Anna Korhonen, and Barbara Plank. 2025. [A rose by any other name: LLM-generated explanations are good proxies for human explanations to collect label distributions on NLI](#). In *Findings of the Association for Computational Linguistics: ACL 2025*.

Beiduo Chen, Xinpeng Wang, Siyao Peng, Robert Litschko, Anna Korhonen, and Barbara Plank. 2024. [“Seeing the big through the small”: Can LLMs approximate human judgment distributions on NLI from a few explanations?](#) In *Findings of the Association for Computational Linguistics: EMNLP 2024*.

- Giulia DeSalvo, Christopher Mohri, Mehryar Mohri, and Yutao Zhong. 2025. [Budgeted multiple-expert deferral](#). *arXiv preprint arXiv:2510.26706*.
- Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. 2023. [ChatGPT outperforms crowd workers for text-annotation tasks](#). *Proceedings of the National Academy of Sciences*, 120(30).
- Kristina Gligorić, Tijana Zrnica, Cinoo Lee, Emmanuel Candès, and Dan Jurafsky. 2025. [Can unconfident LLM annotations be used for confident conclusions?](#) In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*.
- Cornelia Gruber, Helen Alber, Bernd Bischl, Göran Kauermann, Barbara Plank, and Matthias Aßenmacher. 2025. [Revisiting active learning under \(human\) label variation](#). *arXiv preprint arXiv:2507.02593*.
- Ahmad Dawar Hakimi, Ludwig Hirlimann, Isabelle Augenstein, and Hinrich Schütze. 2026. [Do we still need humans in the loop? comparing human and LLM annotation in active learning for hostility detection](#). *arXiv preprint arXiv:2604.13899*.
- Chaemin Jang, Dongman Lee, and Jihee Kim. 2026. [Instruction-tuned language models cannot sample from distributions they can describe](#). *arXiv preprint arXiv:2607.25292*. To appear in Findings of EMNLP 2026.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, et al. 2022. [Language models \(mostly\) know what they know](#). *arXiv preprint arXiv:2207.05221*.
- Hannah Kim, Kushan Mitra, Rafael Li Chen, Sajjadur Rahman, and Dan Zhang. 2024. [MEGAnno+: A human-LLM collaborative annotation system](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*.
- Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. 2024. [Understanding the effects of RLHF on LLM generalisation and diversity](#). In *International Conference on Learning Representations (ICLR)*.
- Christopher Klugmann, Rafid Mahmood, Guruprasad Hegde, Amit Kale, and Daniel Kondermann. 2024. [No need to sacrifice data quality for quantity: Crowd-informed machine annotation for cost-effective understanding of visual data](#). *arXiv preprint arXiv:2409.00048*.
- Guneet Kohli. 2026. [Metric-dependent annotation saturation for learning from label distributions](#). *arXiv preprint arXiv:2605.29797*.
- Artur Kulmizev, Erika Lombart, Patrick Watrin, and Marie-Catherine de Marneffe. 2026. [Label and explanation variation in LLM-based annotation: a case study in natural language inference](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Ryan Lail. 2026. [Decomposing LLM-judge uncertainty to target expert labels](#). *arXiv preprint arXiv:2609.06444*.
- Noah Lee, Na Min An, and James Thorne. 2023. [Can large language models capture dissenting human voices?](#) In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Elisa Leonardelli et al. 2025. [The third edition of the learning with disagreements shared task \(LeWiDi-2025\)](#). In *Proceedings of the 4th Workshop on Perspectivist Approaches to NLP (NLPerspectives)*.
- Minzhi Li, Taiwei Shi, Caleb Ziems, Min-Yen Kan, Nancy F. Chen, Zhengyuan Liu, and Diyi Yang. 2023. [CoAnnotating: Uncertainty-guided work allocation between human and large language models for data annotation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. [Teaching models to express their uncertainty in words](#). *Transactions on Machine Learning Research*.
- Lovish Madaan, David Esiobu, Pontus Stenetorp, Barbara Plank, and Dieuwke Hupkes. 2025. [Lost in inference: Rediscovering the role of natural language inference for large language models](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*.
- David Madras, Toniann Pitassi, and Richard Zemel. 2018. [Predict responsibly: Improving fairness and accuracy by learning to defer](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Nupoor Mehrotra, Adam Visokay, and Kristina Gligorić. 2026. [Multi-perspective LLM annotations for valid analyses in subjective tasks](#). *arXiv preprint arXiv:2603.21404*.
- Nicole Meister, Carlos Guestrin, and Tatsunori Hashimoto. 2025. [Benchmarking distributional alignment of large language models](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*.
- Kei Moriyama, Kouta Nakayama, and Yukino Baba. 2026. [Task assignment meets annotator modeling: Human-LLM collaborative annotation with constraints](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics: Student Research Workshop*.

- Hussein Mozannar and David Sontag. 2020. [Consistent estimators for learning to defer to an expert](#). In *Proceedings of the 37th International Conference on Machine Learning (ICML)*.
- Jingwei Ni, Yu Fan, Vilém Zouhar, Donya Rooein, Alexander Hoyle, Mrinmaya Sachan, Markus Leipold, Dirk Hovy, and Elliott Ash. 2026. [Can reasoning help large language models capture human annotator disagreement?](#) In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.
- Yixin Nie, Xiang Zhou, and Mohit Bansal. 2020. [What can we learn from collective human opinions on natural language inference data?](#) In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9131–9143.
- Ellie Pavlick and Tom Kwiatkowski. 2019. [Inherent disagreements in human textual inferences](#). *Transactions of the Association for Computational Linguistics*, 7:677–694.
- Charlotte Peale, Siddhartha Devic, Parikshit Gopalan, Udi Wieder, and Aravind Gollakota. 2026. [Flexible routing via uncertainty decomposition](#). *arXiv preprint arXiv:2605.07805*.
- Barbara Plank. 2022. [The “problem” of human label variation: On ground truth in data, modeling and evaluation](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Antonio Purificato, Maria Sofia Bucarelli, Andrea Bacciu, Amin Mantrach, and Fabrizio Silvestri. 2026. [QUORUM: QUALity-optimized routing using multiple annotators](#). *arXiv preprint arXiv:2608.27974*.
- Michael V. Reiss. 2023. [Testing the reliability of ChatGPT for text annotation and classification: A cautionary remark](#). *arXiv preprint arXiv:2304.11085*.
- Hope Schroeder, Deb Roy, and Jad Kabbara. 2025. [Just put a human in the loop? investigating LLM-assisted annotation for subjective tasks](#). In *Findings of the Association for Computational Linguistics: ACL 2025*.
- Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. 2023. [Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Petter Törnberg. 2023. [ChatGPT-4 outperforms experts and crowd workers in annotating political Twitter messages with zero-shot learning](#). *arXiv preprint arXiv:2304.06588*.
- Alexandra N. Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. [Learning from disagreement: A survey](#). *Journal of Artificial Intelligence Research*, 72:1385–1470.
- Xinpeng Wang, Bolei Ma, Chengzhi Hu, Leon Weber-Genzel, Paul Röttger, Frauke Kreuter, Dirk Hovy, and Barbara Plank. 2024. [“My answer is C”: First-token probabilities do not match text answers in instruction-tuned language models](#). In *Findings of the Association for Computational Linguistics: ACL 2024*.
- Leon Weber-Genzel, Siyao Peng, Marie-Catherine de Marneffe, and Barbara Plank. 2024. [VariErr NLI: Separating annotation error from human label variation](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Jingyao Wu, Ashley Wang, Keane Ong, Paul Pu Liang, and Rosalind Picard. 2026. [SHALA-LLM: Smartly handling ambiguous labels in aligning LLMs](#). *arXiv preprint arXiv:2606.05376*.
- Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. 2024. [Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs](#). In *International Conference on Learning Representations (ICLR)*.
- Hao Zeng, Huipeng Huang, Xinhao Qu, Jianguo Huang, Bingyi Jing, and Hongxin Wei. 2026. [HyPAC: Cost-efficient LLMs-human hybrid annotation with PAC error guarantees](#). *arXiv preprint arXiv:2602.02550*.
- Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. 2026. [Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity](#). In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.
- Xiang Zhou, Yixin Nie, and Mohit Bansal. 2022. [Distributed NLI: Learning to predict human opinion distributions for language reasoning](#). In *Findings of the Association for Computational Linguistics: ACL 2022*.